

MIDTERM EXAM

ECO403/503–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

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1. Akbobek, Bayan, and Uldana meet at a party to which economists and sociologists have been invited. Every attendee at the party is either an economist or a sociologist (but not both). Every economist knows all the other economists, but every sociologist does not know any of the other attendees, whether they are economists or sociologists. The goal of this question is to analyze what do Akbobek, Bayan, and Uldana know about each other's professions.
 - (a) Provide a knowledge space, using only eight states of the world, that describes the situation above. Describe the knowledge space in terms of its equivalent descriptions using:
 - i. Type spaces and signalling functions.
 - ii. Information partitions.
 - (b) What are the atoms of the common knowledge algebra?
 - (c) At what states of the world is the event "There is at least one economist" common knowledge?
 - (d) At what states of the world is the event "There is at least one sociologist" common knowledge?
 - (e) Suppose that Akbobek and Uldana are economists and that Bayan is a sociologist. Show that it is common knowledge between Akbobek and Uldana that the two of them are economists and Bayan is a sociologist, but this fact is not common knowledge among the three of them.

2. Suppose that Mukhamejan (Player 2) owns an oil field in Atyrau. He expects the profit from the oil field to be negative, and therefore intends to sell it. Adiletkhan (Player 1) is of the opinion that the oil field can yield positive profits, and is therefore willing to purchase it (for simplicity, at the price of \$0). Adiletkhan and Mukhamejan arrive at different determinations because they have different information. Suppose the following exchange happens:

- Adiletkhan: I am interested in purchasing the oil field; are you interested in selling?
- Mukhamejan: Yes, I am interested in selling; are you interested in purchasing?
- Adiletkhan: Yes, I am interested in purchasing; are you still interested in selling?
- And so on, until one of the two parties announces that he has no interest in a deal.

You will model the situation using the following belief space:

$$N = \{1, 2\}, \quad \Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\mathcal{P}_1 = \{\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}\}, \quad \mathcal{P}_2 = \{\{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9\}\}$$

$$\mu(\omega) = \frac{1}{9}, \quad \forall \omega \in \Omega.$$

The states in Ω represent all variables affecting the profit of the oil field. Consider the random variable $X : \Omega \rightarrow \mathbb{R}$, given by:

$$X(\omega) = \begin{cases} +1 & \omega \in \{1, 2, 5, 6, 9\}, \\ -1 & \omega \in \{3, 4, 7, 8\}. \end{cases}$$

$X(\omega)$ represents the profit, in billions of tenge, yielded by the oil field at the state of the world ω . At each stage, Adiletkhan will be interested in the deal only if the conditional expectation of X given his information is positive, and Mukhamejan will be interested in the deal only if the conditional expectation of X given his information is negative. The process looks like this:

- Player 1 states whether or not $E[X|\Sigma_1](\omega) > 0$ (implicitly doing so by expressing or not expressing interest in purchasing the oil field). If he says “no” (i.e., his expectation is less than or equal to 0), the process ends here.
- If the process gets to the second stage, player 2 states whether his expectation of X , given the information he has received so far, is negative or not. The information he has includes $P_2(\omega) \in \Sigma_2$, and the affirmative interest of Player 1 in the first stage. If player 2 now says “no” (i.e., his expectation is greater than or equal to 0), the process ends here.
- If the process has not yet ended, player 1 states whether his expectation of X , given the information he has received so far, is positive or not. The information he has includes $P_1(\omega) \in \Sigma_1$, and the affirmative interest of player 2 in the second stage. If player 1 now says “no” (i.e., his expectation is less than or equal to 0), the process ends here.
- And so on. The process ends the first time either Player 1’s expectation of X , given his information, is not positive, or Player 2’s expectation of X , given his information, is not negative.

- How does the process look like at state of the world $\omega = 5$?
- How does the process look like at state of the world $\omega = 1$?
- How does the process look like at state of the world $\omega = 4$?
- How does the process look like at state of the world $\omega = 8$?
- How does the process look like at state of the world $\omega = 9$?

3. *The market for lemons.* Consider a used car market in which a fraction q of the cars ($0 \leq q \leq 1$) are in good condition and $1 - q$ are in bad conditions (lemons). The seller (Player 2) knows the quality of the car she is offering to sell while the buyer (Player 1) does not know the quality of the car that he is being offered to buy. Each used car is offered for a sale at the price of $\$p$ (in units of million tenge). The payoffs to the seller and the buyer, depending on whether or not the transaction is completed, are described in the following tables:

		Seller	
		Sell	Don't Sell
Buyer	Buy	$6 - p, p$	$0, 5$
	Don't Buy	$0, 5$	$0, 5$

State game if car in good condition

		Seller	
		Sell	Don't Sell
Buyer	Buy	$4 - p, p$	$0, 0$
	Don't Buy	$0, 0$	$0, 0$

State game if car in bad condition

- Provide the belief space representing the Bayesian Game described above.
- Describe the set of *pure strategies* for each of the players.
- What happens in the extreme cases of $q = 0$ and $q = 1$? Note that in these two cases there is *complete information* about the quality.
- Assume $q \in (0, 1)$. For what parameter values p and q does the seller *always* wants to sell, while the buyer never wants to buy in equilibrium?¹ *Hint: if you cannot think about this task with general parameters p and q , try to find specific values for the parameters such that the statement of the task is true.*
- For each pair of parameters p and q , *find all* the Bayesian Nash Equilibria.

¹This another example of adverse selection against the quality of the cars. For these parameter values, we say that the *market unravels*: no buyer wants to buy a car of good quality, even if they would be willing to do so if they knew the quality, and thus, there is a complete collapse of the market.